

Investigating Double-Slit Interference

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Abstract—This experiment investigated far-field diffraction from a double slit using laser light, with the aim of determining the slit separation from the interference pattern formed in the focal plane of a lens. The positions of intensity minima were measured using a photodiode on a translation stage, and the full intensity distribution was analysed using a CMOS camera.

The minima-spacing method gave a slit separation of $d = (0.213 \pm 0.010)$ mm, while fitting the theoretical double-slit intensity distribution to the CMOS data yielded $d = (0.2406 \pm 0.0003)$ mm and a slit width of $a = (0.0665 \pm 0.0004)$ mm. Although the CMOS method provided much higher precision, the two values of d differ beyond their quoted uncertainties, indicating systematic effects and the need for further analysis. Overall, the observed diffraction pattern was well described by the theoretical model.

I. INTRODUCTION

Diffraction and interference both arise from the superposition of waves and the resulting variations in intensity that depend on the relative phases of the contributing wavefronts. In optics, diffraction refers to the patterns produced by light passing through apertures of finite width whereas interference refers to patterns formed by a small number of distinct sources (e.g light from two narrow slits).

In this experiment, diffraction patterns produced by single and double slits are investigated using laser light in the far-field regime. Rather than observing diffraction patterns at a large distance from the slits, the far-field regime is achieved by always making observations at the focal plane of a converging lens [3].

The primary focus of this experiment is the analysis of the double-slit interference pattern, from which the slit-separation is determined. Positions of minimum intensity are measured and then related to the geometry of the double-slit setup under a small angle approximation. Furthermore, images of the interference patterns produced are analysed and fitted to a theoretical intensity distribution to more accurately determine slit-separation.

II. THEORY

When coherent, monochromatic light passes through two parallel slits, constructive and destructive interference occur depending on the path difference between light coming from both slits. In the far-field regime, the path difference δ between light from the two slits arriving at angle θ is given by

$$\delta = d \sin \theta$$

where d is the slit-separation. This can be seen through Figure 1. Destructive interference (points of intensity minima) occur when [2]

$$d \sin \theta = (m + \frac{1}{2})\lambda$$

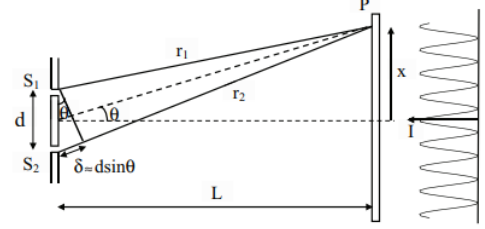


Fig. 1. Figure from [3]. Depicts the geometry for the double-slit interference system: For large L , the path difference to point P is about $d \sin \theta$, where d is the slit-separation. The effects of large L are achieved by making observations at the focal plane of a lens in this experiment.

where $m = 0, \pm 1, \pm 2, \dots$ and λ is the wavelength of the light.

The experimental setup used did not use large L as shown in Figure 1, instead observations were made at the focal plane of a lens of focal distance f (which has an equivalent effect). Under the small-angle approximation, the relationship between the distance of a point x along the pattern and θ is

$$\theta \approx \frac{x}{f}$$

Combining this with the destructive interference condition yields the positions of intensity minima which is given by

$$x_m = (m + \frac{1}{2}) \frac{\lambda f}{d} \quad (1)$$

Equation 1 thus predicts a linear relationship between minima number m and position x . By experimentally measuring the positions of successive minima, the slit-separation d can be obtained through the gradient of a linear fit between x_m and m .

In practice, both slits have a width a , which modifies the intensity distribution through an envelope of a single-slit diffraction pattern. The resulting intensity distribution is then given by

$$I(x) = 4I_0 \left[\frac{\sin\left(\frac{\pi a x}{\lambda f}\right)}{\frac{\pi a x}{\lambda f}} \right]^2 \cos^2\left(\frac{\pi d x}{\lambda f}\right) + k \quad (2)$$

where I_0 is the intensity due to a single slit and k is a constant representing the background intensity. Equation 2 is used as a fit for data from digital images of the double-slit interference pattern obtained with a CMOS camera.

The above equations make two key assumptions: that the diffraction observed followed the far-field regime and that the small angle approximation held. Deviations from these assumptions as well as other experimental factors, such as finite detector resolution, contribute to the uncertainties in the measured slit-separation.

III. METHODS

Figure 2 describes the experimental setup. A monochromatic laser ($\lambda = 670 \pm 1\text{nm}$) was directed onto the double-slit diffracting object mounted onto an optical bench. A lens of focal length $f = 500\text{mm}$ was placed in front of the diffracting object. All observations were taken at $f = 500\text{mm}$ away from the lens, measured using the track on the optical bench. f 's measurement uncertainty thus comes the resolution of the track making $f = (500 \pm 0.5)\text{mm}$.

Positions of intensity minima were determined using a photodiode mounter behind a $100\mu\text{m}$ slit, connected to a voltmeter and attached to a precision translation stage. The detector was translated across the diffraction pattern, and the position corresponding to a minimum in the detected signal (a minimum voltage) was recorded. Only the positions of these minima were measured, rather than the full intensity profile. The uncertainty of these measurements comes partially from the measurement uncertainty of the precision stage ($\pm 0.005\text{mm}$). The dominant factor that affects the uncertainty in the use of the photodiode is the finite width of the slit in front of it. Because the detector measures the average intensity over this width, minima appear broadened and their exact positions are less sharply defined. Half the width of this slit ($\pm 50\mu\text{m}$) is taken as the contribution of this effect to the uncertainty. The precision of the stage and half the slit width are combined in quadrature to estimate the positional uncertainty of a given minima ($\pm 0.0502\text{mm}$).

To obtain an image of the intensity distribution, the photodiode was then replaced with a CMOS camera positioned at the focal plane of the same focal length lens ($f = 500\text{mm}$). This CMOS camera had a sensor of dimensions $6.66\text{mm} \times 5.32\text{mm}$ (H x V) with a resolution of 1280×1024 pixels. The exposure time of the camera was reduced and polarising filters were used in order to limit saturation of the detector. Images of the double slit interference pattern were then recorded. Each image was rotated such that fringes were aligned horizontally. An intensity profile of across the central region of the pattern was extracted using image analysis software. Pixel positions were converted to physical distances using the dimensions of the camera above. An estimate of the uncertainty in the distances found using the CMOS camera was taken to be the width one pixel spans, about $\pm 0.005\text{mm}$.

For the double-slit photo diode data, the measured minima positions x_m were plotted against the corresponding minima number (determined by proximity to the central maxima). A linear fit was then performed using the python module `numpy.polyfit`. Using Equation (1), the slit-separation d is given by

$$d = \frac{\lambda f}{G}$$

where G is the gradient of the fitted line. The uncertainty in d was then calculated by propagating the uncertainty in the fitted gradient (obtained through the square root of the diagonals of the covariance matrix outputted by `numpy.polyfit`) with the uncertainties in λ and f . This was achieved through adding the relative uncertainties in quadrature.

For the CMOS data, the measured intensity profile was fitted to Equation (2), the theoretical double-slit intensity

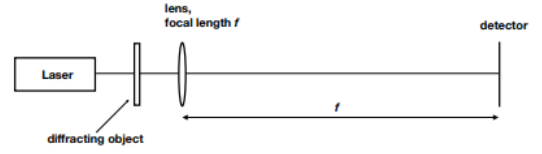


Fig. 2. Figure from [3]. Experimental Setup used for the measurement of interference and diffraction patterns. The diffracting object (double slits) is placed as close to the laser, the detector is placed at the focal plane of the lens.

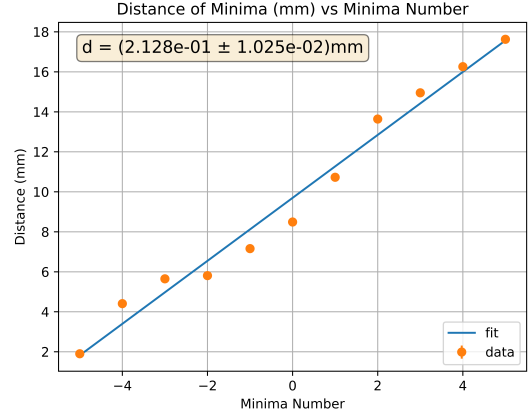


Fig. 3. Graph showing x_m versus m (data collected using the photodiode). A weighted linear fit was applied to the data, according to Equation 1.

distribution, with slit-separation d and slit width a as free parameters. This was done using the python module `scipy.optimize.curve_fit`. The uncertainty in these parameters is given by the square root of the diagonals of the covariance matrix outputted by the fitting module.

IV. RESULTS & DISCUSSION

The data measured using the photodiode and the precision stage are presented in Figure 3 which depicts a plot of minima position x_m versus minima number m . The uncertainty in each minima position was estimated as $\pm 0.05\text{mm}$, an addition in quadrature of the effect of the finite width of the slit placed in front of the photodiode ($\pm 50\mu\text{m}$) and the precision of the track ($\pm 0.005\text{mm}$). The fitted gradient obtained in Figure 3 is below:

$$G = (1.57 \pm 0.076)\text{mm}$$

Using Equation (1) and this gradient, the value for d and its uncertainty can be determined (for the minima spacing method):

$$d = (0.213 \pm 0.010)\text{mm}$$

The uncertainty in the diode value for d comes from the combination of uncertainties in the wavelength λ , focal length f and fitted gradient G , combined in quadrature as described in Section III. The fractional value of this uncertainty is around 4.7%.

The Intensity profile of the double-slit interference pattern along with a fit using Equation (2) is presented in Figure 4.

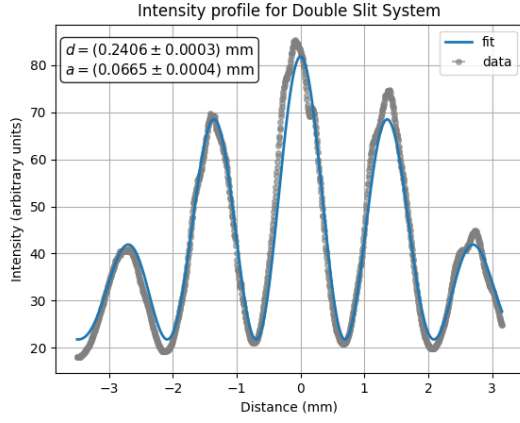


Fig. 4. Graph showing Equation (2) fitted to the intensity profile of the double-slit interference pattern (obtained using the CMOS camera).

The fit behaves expected, having even fringe spacing and a diffraction envelope. The fit parameters were

$$d = (0.2406 \pm 0.0003) \text{ mm}$$

$$a = (0.0665 \pm 0.0004) \text{ mm}$$

The slit separation obtained using the photodiode minima spacing method ($d = (0.213 \pm 0.010) \text{ mm}$) differs from the value obtained using the CMOS intensity profile fitting method ($d = (0.2406 \pm 0.0003)$). Although the CMOS method has much higher precision, the difference between the two values is quite significant:

$$\frac{\text{Difference}}{\text{Combined Uncertainty}} = \frac{|0.2406 - 0.213|}{\sqrt{0.010^2 + 0.0003^2}} \approx 2.8\sigma,$$

The two values vary by about 2.8σ which means that the discrepancy is unlikely due to their quoted uncertainties. This is indicative of systematic effects in one of or both of the methodologies used.

For the photodiode method, it is likely that the majority of systematic uncertainty arises from the presence of the $100\mu\text{m}$ slit in front of the detector. This slit causes the signal measured to represent an average over some finite spatial width. This means that the exact positions of minima are less sharply defined. This effect likely becomes more significant for higher order minima, as their intensities are lower. The uncertainty in the precision stage $\pm 0.005 \text{ mm}$, however, would not have a significant impact. The estimation of the uncertainty due to the slit width $\pm 50\mu\text{m}$ dominates the combination of these two uncertainties. The uncertainty in the fitted gradient (and hence the uncertainty in the photodiode value for d) is more of a reflection of the difficulty of locating minima, rather than that of the mechanical precision of the stage.

The CMOS method, in contrast, benefits from a much finer spatial resolution. Assuming that a given feature (minima) can be located up to about 1 pixel, the uncertainty in any given position for this method is about $\pm 0.005 \text{ mm}$. Consequently, it is likely that any systematic uncertainty in this method comes from other aspects of the method such as the model (Equation 2) used to fit the intensity distribution. This model

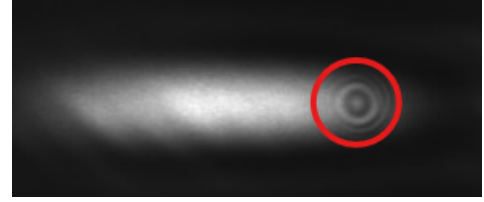


Fig. 5. Portion of image used to generate the intensity profile with optical aberration circled in red.

has several key assumptions that may not hold in reality. It assumes that angles are small and that all diffraction is of the far-field regime, for example. Any deviations from these conditions may lead to discrepancies between the measured intensity profile and the fitted equation which increase the systematic uncertainties in this method.

Another aspect of the CMOS method which may increase the systematic uncertainty is the unavoidable presence of optical aberrations in the images used to generate the intensity profiles. Figure 5 shows part of the image used to generate the intensity profile. Circled in red is an optical aberration which likely led to the measurement of an intensity profile which Equation (2) can not model accurately. This impact can potentially be seen in Figure 4. The right maxima adjacent to the central maximum deviates at the peak compared to the model. This deviation could be due to the presence of an optical aberration (not necessarily the one depicted in Figure 5). The presence of optical aberrations likely increases the systematic uncertainty in the CMOS camera method and hence the value of slit-separation obtained.

V. CONCLUSION

This experiment investigated far-field diffraction from a double-slit aperture and aimed to determine the slit separation from the observed interference pattern. Using the minima-spacing method with a photodiode, the slit separation was found to be $d = (0.213 \pm 0.010) \text{ mm}$. Fitting the full intensity distribution recorded using a CMOS camera yielded $d = (0.2406 \pm 0.0003) \text{ mm}$ and a slit width of $a = (0.0665 \pm 0.0004) \text{ mm}$. The CMOS method provided higher precision due to its finer spatial resolution; however, the two values of d differed beyond their combined uncertainties, indicating the presence of systematic effects. The primary systematic effects involved were likely the width of the slit in front of the diode and limitations of the theoretical intensity model for the photodiode and CMOS method respectively. Despite systematic discrepancies, both methods produced slit separations of similar magnitude and the observed interference pattern closely followed theoretical predictions.

REFERENCES

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